

Министерство науки и высшего образования Российской Федерации
федеральное государственное автономное образовательное учреждение высшего
образования
«НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ УНИВЕРСИТЕТ ИТМО»

Отчет

по Лабораторной работе №5

по дисциплине «Практика и Реализация Баз Данных»

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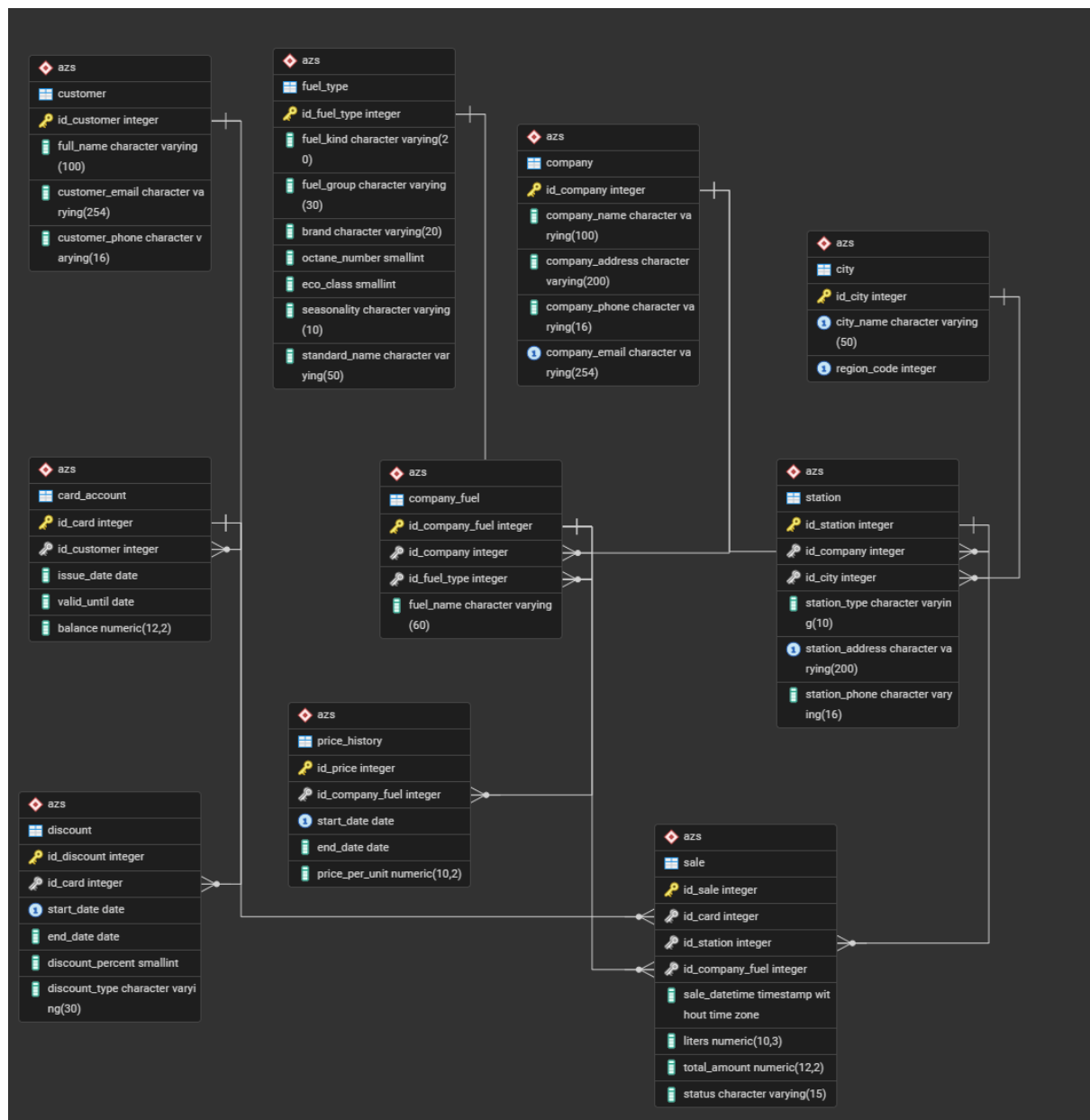


Цель работы:

Овладеть практическими создания и использования процедур, функций и триггеров в базе данных PostgreSQL.

Выбранный вариант - Вариант 2.1.

1. Схема базы данных



2. Подключаемся к базе

```
SQL Shell (psql)
Server [localhost]:
Database [postgres]:
Port [5432]:
Username [postgres]:
Password for user postgres:

psql (18.3)
WARNING: Console code page (866) differs from Windows code page (1251)
        8-bit characters might not work correctly. See psql reference
        page "Notes for Windows users" for details.
Type "help" for help.

postgres=# \c azs_sales_db
You are now connected to database "azs_sales_db" as user "postgres".
azs_sales_db=# \! chcp 1251
Active code page: 1251
azs_sales_db=# set search_path to azs, public;
SET
azs_sales_db=# \dt
          List of tables
Schema |      Name      | Type  | Owner
-----+-----+-----+-----
 azs   | card_account   | table | postgres
 azs   | city           | table | postgres
 azs   | company        | table | postgres
 azs   | company_fuel   | table | postgres
 azs   | customer       | table | postgres
 azs   | discount       | table | postgres
 azs   | fuel_type      | table | postgres
 azs   | price_history  | table | postgres
 azs   | sale           | table | postgres
 azs   | station        | table | postgres
(10 rows)
```

3. Скрипт процедур/функций и триггеров

Скрипт находится в той же папке что и отчёт, lab5_create.sql
Запуск скрипта:

```

azs_sales_db=# \i 'C:/Users/UserPC/Desktop/yche6a/DB/DATABASES_PIn_09.03.03_2025-2026/students/K3240/Konanchuk_Ivan_Alekseevich/LR_5/lab5.sql'
SET
CREATE FUNCTION
CREATE FUNCTION
CREATE PROCEDURE
CREATE FUNCTION
CREATE FUNCTION
CREATE FUNCTION
CREATE FUNCTION
CREATE FUNCTION
CREATE FUNCTION
CREATE FUNCTION
psql:C:/Users/UserPC/Desktop/yche6a/DB/DATABASES_PIn_09.03.03_2025-2026/students/K3240/Konanchuk_Ivan_Alekseevich/LR_5/lab5.sql:336: NOTICE: trigger "trg_01_sale_check_card_validity" for relation "sale" does not exist, skipping
DROP TRIGGER
psql:C:/Users/UserPC/Desktop/yche6a/DB/DATABASES_PIn_09.03.03_2025-2026/students/K3240/Konanchuk_Ivan_Alekseevich/LR_5/lab5.sql:337: NOTICE: trigger "trg_02_sale_check_station_fuel_company" for relation "sale" does not exist, skipping
DROP TRIGGER
psql:C:/Users/UserPC/Desktop/yche6a/DB/DATABASES_PIn_09.03.03_2025-2026/students/K3240/Konanchuk_Ivan_Alekseevich/LR_5/lab5.sql:338: NOTICE: trigger "trg_03_sale_calculate_total_amount" for relation "sale" does not exist, skipping
DROP TRIGGER
psql:C:/Users/UserPC/Desktop/yche6a/DB/DATABASES_PIn_09.03.03_2025-2026/students/K3240/Konanchuk_Ivan_Alekseevich/LR_5/lab5.sql:339: NOTICE: trigger "trg_04_sale_decrease_card_balance" for relation "sale" does not exist, skipping
DROP TRIGGER
psql:C:/Users/UserPC/Desktop/yche6a/DB/DATABASES_PIn_09.03.03_2025-2026/students/K3240/Konanchuk_Ivan_Alekseevich/LR_5/lab5.sql:340: NOTICE: trigger "trg_05_sale_restore_balance_on_cancel" for relation "sale" does not exist, skipping
DROP TRIGGER
psql:C:/Users/UserPC/Desktop/yche6a/DB/DATABASES_PIn_09.03.03_2025-2026/students/K3240/Konanchuk_Ivan_Alekseevich/LR_5/lab5.sql:341: NOTICE: trigger "trg_06_price_history_close_previous" for relation "price_history" does not exist, skipping
DROP TRIGGER
psql:C:/Users/UserPC/Desktop/yche6a/DB/DATABASES_PIn_09.03.03_2025-2026/students/K3240/Konanchuk_Ivan_Alekseevich/LR_5/lab5.sql:342: NOTICE: trigger "trg_07_discount_close_previous" for relation "discount" does not exist, skipping
DROP TRIGGER
CREATE TRIGGER
CREATE TRIGGER
CREATE TRIGGER
CREATE TRIGGER
CREATE TRIGGER
CREATE TRIGGER
CREATE TRIGGER

```

Всё работает правильно, при первом запуске скрипт безуспешно пытается удалить еще не существующие триггеры.

Смотрим созданные триггеры:

```

azs_sales_db=# select
azs_sales_db=#     event_object_table as table_name,
azs_sales_db=#     trigger_name,
azs_sales_db=#     action_timing,
azs_sales_db=#     event_manipulation
azs_sales_db=# from information_schema.triggers
azs_sales_db=# where trigger_schema = 'azs'
azs_sales_db=# order by event_object_table, trigger_name;

```

table_name	trigger_name	action_timing	event_manipulation
discount	trg_07_discount_close_previous	BEFORE	INSERT
price_history	trg_06_price_history_close_previous	BEFORE	INSERT
sale	trg_01_sale_check_card_validity	BEFORE	UPDATE
sale	trg_01_sale_check_card_validity	BEFORE	INSERT
sale	trg_02_sale_check_station_fuel_company	BEFORE	INSERT
sale	trg_02_sale_check_station_fuel_company	BEFORE	UPDATE
sale	trg_03_sale_calculate_total_amount	BEFORE	UPDATE
sale	trg_03_sale_calculate_total_amount	BEFORE	INSERT
sale	trg_04_sale_decrease_card_balance	AFTER	INSERT
sale	trg_05_sale_restore_balance_on_cancel	AFTER	UPDATE

(10 rows)

4. Проверки по индивидуальному заданию 4

1. Покупки клиента за заданную дату:

```

azs_sales_db=# select *
azs_sales_db=# from fn_customer_purchases_by_date(1, date '2026-03-05')
azs_sales_db=# limit 10;

```

	customer_id	customer_name	customer_email	customer_phone	sale_date	sale_time	station_id	fuel_name	liters	paid_amount
	1	Ivan Ivanov	ivanov@test.ru	+79000000001	2026-03-05	10:15:00	1	AI-		
92 Lukoil	30.500	1677.65								
	1	Ivan Ivanov	ivanov@test.ru	+79000000001	2026-03-05	10:15:00	1	AI-		
92 Lukoil	30.500	1677.65								
	1	Ivan Ivanov	ivanov@test.ru	+79000000001	2026-03-05	10:15:00	1	AI-		
92 Lukoil	30.500	1677.65								
	1	Ivan Ivanov	ivanov@test.ru	+79000000001	2026-03-05	10:15:00	1	AI-		
92 Lukoil	30.500	1677.65								
	1	Ivan Ivanov	ivanov@test.ru	+79000000001	2026-03-05	10:15:00	1	AI-		
92 Lukoil	30.500	1677.65								
	1	Ivan Ivanov	ivanov@test.ru	+79000000001	2026-03-05	10:15:00	1	AI-		
92 Lukoil	30.500	1677.65								
	1	Ivan Ivanov	ivanov@test.ru	+79000000001	2026-03-05	10:15:00	1	AI-		
92 Lukoil	30.500	1677.65								

(10 rows)

2. Количество видов топлива по каждой фирме:

```

azs_sales_db=# select *
azs_sales_db=# from fn_company_fuel_type_count();

```

company_id	company_name	fuel_type_count
1	Lukoil	2
2	Gazprom Neft	2

(2 rows)

3. Добавление новой тестовой АЗГС

```
azs_sales_db=# begin;
BEGIN
azs_sales_db=#
azs_sales_db=# call pr_add_azgs_station(
azs_sales_db(##      1,
azs_sales_db(##      1,
azs_sales_db(##      'Saint Petersburg, Lab5 AZGS, 1',
azs_sales_db(##      '+78125550115',
azs_sales_db(##      null::integer
azs_sales_db(## );
p_new_station_id
-----
5
(1 row)

azs_sales_db=#
azs_sales_db=# select *
azs_sales_db=# from station
azs_sales_db=# order by id_station desc
azs_sales_db=# limit 3;
 id_station | id_company | id_city | station_type | station_address | station_phone
-----
5 | 1 | 1 | AZGS | Saint Petersburg, Lab5 AZGS, 1 | +78125550115
4 | 2 | 3 | AZGS | Kazan, Pobedy Ave, 40 | +78435550004
4 | 2 | 3 | AZGS | Kazan, Pobedy Ave, 40 | +78435550004
(3 rows)

azs_sales_db=#
azs_sales_db=# rollback;
ROLLBACK
azs_sales_db=#
```

5. Проверка триггеров

1. Проверка автоподсчета суммы и списания баланса. Триггеры:
 - a. trg_01_sale_check_card_validity
 - b. trg_02_sale_check_station_fuel_company
 - c. trg_03_sale_calculate_total_amount
 - d. trg_04_sale_decrease_card_balance

```

azs_sales_db=# select id_card, balance
azs_sales_db=# from card_account
azs_sales_db=# where id_card = 1;
 id_card | balance
-----+-----
      1 | 5000.00
      1 | 5000.00
(2 rows)

azs_sales_db=#
azs_sales_db=# insert into sale (
azs_sales_db=#     id_sale,
azs_sales_db=#     id_card,
azs_sales_db=#     id_station,
azs_sales_db=#     id_company_fuel,
azs_sales_db=#     sale_datetime,
azs_sales_db=#     liters,
azs_sales_db=#     total_amount,
azs_sales_db=#     status
azs_sales_db=# )
azs_sales_db=# values (
azs_sales_db=#     900001,
azs_sales_db=#     1,
azs_sales_db=#     1,
azs_sales_db=#     1,
azs_sales_db=#     now(),
azs_sales_db=#     10.000,
azs_sales_db=#     0,
azs_sales_db=#     'completed'
azs_sales_db=# );
INSERT 0 1
azs_sales_db=# select
azs_sales_db=#     id_sale,
azs_sales_db=#     id_card,
azs_sales_db=#     id_station,
azs_sales_db=#     id_company_fuel,
azs_sales_db=#     sale_datetime,
azs_sales_db=#     liters,
azs_sales_db=#     total_amount,
azs_sales_db=#     status
azs_sales_db=# from sale
azs_sales_db=# where id_sale = 900001;
 id_sale | id_card | id_station | id_company_fuel | sale_datetime           | liters | total_amount | status
-----+-----+-----+-----+-----+-----+-----+-----
 900001 |      1 |          1 |                1 | 2026-06-01 03:25:58.495817 | 10.000 |       550.05 | completed
(1 row)

azs_sales_db=#
azs_sales_db=# select id_card, balance
azs_sales_db=# from card_account
azs_sales_db=# where id_card = 1;
 id_card | balance
-----+-----
      1 | 4449.95
      1 | 4449.95
(2 rows)

azs_sales_db=#

```

2. Проверка отмены продажи и возврата денег. Триггер trg_05_sale_restore_balance_on_cancel

```

azs_sales_db=# begin;
BEGIN
azs_sales_db=#
azs_sales_db=# select id_card, balance
azs_sales_db=# from card_account
azs_sales_db=# where id_card = 1;
 id_card | balance
-----+-----
      1 | 5000.00
      1 | 5000.00
(2 rows)

```

```

azs_sales_db=# insert into sale (
azs_sales_db=#     id_sale,
azs_sales_db=#     id_card,
azs_sales_db=#     id_station,
azs_sales_db=#     id_company_fuel,
azs_sales_db=#     sale_datetime,
azs_sales_db=#     liters,
azs_sales_db=#     total_amount,
azs_sales_db=#     status
azs_sales_db=# )
azs_sales_db=# values (
azs_sales_db=#     900002,
azs_sales_db=#     1,
azs_sales_db=#     1,
azs_sales_db=#     1,
azs_sales_db=#     now(),
azs_sales_db=#     10.000,
azs_sales_db=#     0,
azs_sales_db=#     'completed'
azs_sales_db=# );
INSERT 0 1
azs_sales_db=#
azs_sales_db=# select id_card, balance
azs_sales_db=# from card_account
azs_sales_db=# where id_card = 1;
 id_card | balance
-----+-----
        1 | 4449.95
        1 | 4449.95
(2 rows)

azs_sales_db=#
azs_sales_db=# update sale
azs_sales_db=# set status = 'cancelled'
azs_sales_db=# where id_sale = 900002;
UPDATE 1
azs_sales_db=#
azs_sales_db=# select id_sale, total_amount, status
azs_sales_db=# from sale
azs_sales_db=# where id_sale = 900002;
 id_sale | total_amount | status
-----+-----+-----
 900002 |      550.05 | cancelled
(1 row)

azs_sales_db=#
azs_sales_db=# select id_card, balance
azs_sales_db=# from card_account
azs_sales_db=# where id_card = 1;
 id_card | balance
-----+-----
        1 | 5000.00
        1 | 5000.00
(2 rows)

```

3. Проверка ошибки при невалидной карте. Триггер
trg_01_sale_check_card_validity


```

azs_sales_db=# begin;
BEGIN
azs_sales_db=# insert into sale (
azs_sales_db(*#      id_sale,
azs_sales_db(*#      id_card,
azs_sales_db(*#      id_station,
azs_sales_db(*#      id_company_fuel,
azs_sales_db(*#      sale_datetime,
azs_sales_db(*#      liters,
azs_sales_db(*#      total_amount,
azs_sales_db(*#      status
azs_sales_db(*# )
azs_sales_db=# values (
azs_sales_db(*#      900003,
azs_sales_db(*#      1,
azs_sales_db(*#      1,
azs_sales_db(*#      1,
azs_sales_db(*#      timestamp '2024-12-01 10:00:00',
azs_sales_db(*#      10.000,
azs_sales_db(*#      0,
azs_sales_db(*#      'completed'
azs_sales_db(*# );
ERROR:  Card 1 is not valid for sale date 2024-12-01
CONTEXT:  PL/pgSQL function fn_trg_sale_check_card_validity() line 17 at RAISE
azs_sales_db=#
azs_sales_db=# rollback;
ROLLBACK

```

4. Проверка ошибки, если станция продает топливо другой фирмы. Триггер `trg_02_sale_check_station_fuel_company`

```

azs_sales_db=# begin;
BEGIN
azs_sales_db=# insert into sale (
azs_sales_db(*#      id_sale,
azs_sales_db(*#      id_card,
azs_sales_db(*#      id_station,
azs_sales_db(*#      id_company_fuel,
azs_sales_db(*#      sale_datetime,
azs_sales_db(*#      liters,
azs_sales_db(*#      total_amount,
azs_sales_db(*#      status
azs_sales_db(*# )
azs_sales_db=# values (
azs_sales_db(*#      900004,
azs_sales_db(*#      1,
azs_sales_db(*#      1,
azs_sales_db(*#      4,
azs_sales_db(*#      now(),
azs_sales_db(*#      10.000,
azs_sales_db(*#      0,
azs_sales_db(*#      'completed'
azs_sales_db(*# );
ERROR:  Fuel 4 belongs to company 2, but station 1 belongs to company 1
CONTEXT:  PL/pgSQL function fn_trg_sale_check_station_fuel_company() line 25 at RAISE
azs_sales_db=#
azs_sales_db=# rollback;
ROLLBACK

```

5. Проверка автоматического закрытия старой цены. Триггер `trg_06_price_history_close_previous`

```

azs_sales_db=# begin;
BEGIN
azs_sales_db=#
azs_sales_db=# select id_price, id_company_fuel, start_date, end_date, price_per_unit
azs_sales_db=# from price_history
azs_sales_db=# where id_company_fuel = 1
azs_sales_db=# order by start_date;
 id_price | id_company_fuel | start_date | end_date | price_per_unit
-----+-----+-----+-----+-----
          1 |          1 | 2026-01-01 | 2026-02-28 |         56.90
          1 |          1 | 2026-01-01 | 2026-02-28 |         56.90
          2 |          1 | 2026-03-01 |          |         57.90
          2 |          1 | 2026-03-01 |          |         57.90
(4 rows)

azs_sales_db=#
azs_sales_db=# insert into price_history (
azs_sales_db=#     id_price,
azs_sales_db=#     id_company_fuel,
azs_sales_db=#     start_date,
azs_sales_db=#     end_date,
azs_sales_db=#     price_per_unit
azs_sales_db=# )
azs_sales_db=# values (
azs_sales_db=#     900001,
azs_sales_db=#     1,
azs_sales_db=#     current_date,
azs_sales_db=#     null,
azs_sales_db=#     60.00
azs_sales_db=# );
INSERT 0 1
azs_sales_db=#
azs_sales_db=# select id_price, id_company_fuel, start_date, end_date, price_per_unit
azs_sales_db=# from price_history
azs_sales_db=# where id_company_fuel = 1
azs_sales_db=# order by start_date;
 id_price | id_company_fuel | start_date | end_date | price_per_unit
-----+-----+-----+-----+-----
          1 |          1 | 2026-01-01 | 2026-02-28 |         56.90
          1 |          1 | 2026-01-01 | 2026-02-28 |         56.90
          2 |          1 | 2026-03-01 | 2026-05-31 |         57.90
          2 |          1 | 2026-03-01 | 2026-05-31 |         57.90
    900001 |          1 | 2026-06-01 |          |         60.00
(5 rows)

azs_sales_db=#
azs_sales_db=# rollback;
ROLLBACK

```

6. Проверка автоматического закрытия старой скидки. Триггер
trg_07_discount_close_previous

```

azs_sales_db=# begin;
BEGIN
azs_sales_db=#
azs_sales_db=# select id_discount, id_card, start_date, end_date, discount_percent, discount_type
azs_sales_db=# from discount
azs_sales_db=# where id_card = 1
azs_sales_db=# order by start_date;
 id_discount | id_card | start_date | end_date | discount_percent | discount_type
-----+-----+-----+-----+-----+-----
          1 |      1 | 2026-01-01 |          |          5 | permanent
          1 |      1 | 2026-01-01 |          |          5 | permanent
(2 rows)

azs_sales_db=#
azs_sales_db=# insert into discount (
azs_sales_db=#     id_discount,
azs_sales_db=#     id_card,
azs_sales_db=#     start_date,
azs_sales_db=#     end_date,
azs_sales_db=#     discount_percent,
azs_sales_db=#     discount_type
azs_sales_db=# )
azs_sales_db=# values (
azs_sales_db=#     900001,
azs_sales_db=#     1,
azs_sales_db=#     current_date,
azs_sales_db=#     null,
azs_sales_db=#     10,
azs_sales_db=#     'lab5_test'
azs_sales_db=# );
INSERT 0 1
azs_sales_db=#
azs_sales_db=# select id_discount, id_card, start_date, end_date, discount_percent, discount_type
azs_sales_db=# from discount
azs_sales_db=# where id_card = 1
azs_sales_db=# order by start_date;
 id_discount | id_card | start_date | end_date | discount_percent | discount_type
-----+-----+-----+-----+-----+-----
          1 |      1 | 2026-01-01 | 2026-05-31 |          5 | permanent
          1 |      1 | 2026-01-01 | 2026-05-31 |          5 | permanent
    900001 |      1 | 2026-06-01 |          |         10 | lab5_test
(3 rows)

azs_sales_db=#
azs_sales_db=# rollback;
ROLLBACK

```

Вывод

В ходе лабораторной работы были созданы функции, процедура и триггеры для базы данных azs_sales_db. Функции позволяют получать сведения о покупках клиента за заданную дату и считать количество видов топлива по фирмам-поставщикам. Процедура используется для добавления новой АЗГС. Также были реализованы триггеры, которые проверяют корректность продажи, автоматически рассчитывают итоговую сумму, изменяют баланс карты, закрывают старые периоды цен и скидок. В результате часть бизнес-логики была перенесена на уровень базы данных.